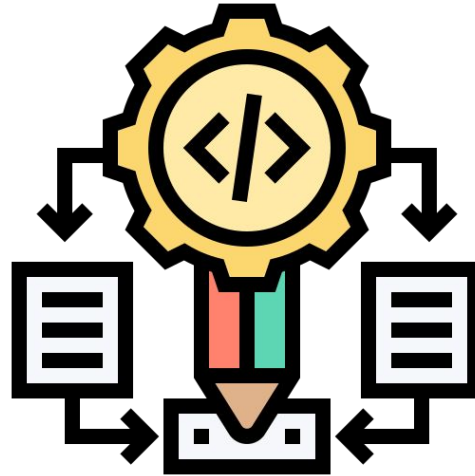


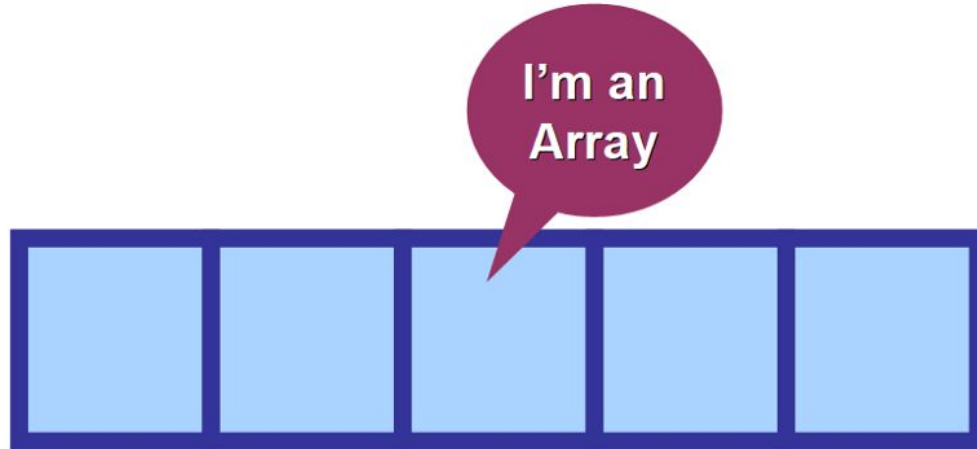
Algorithm 1



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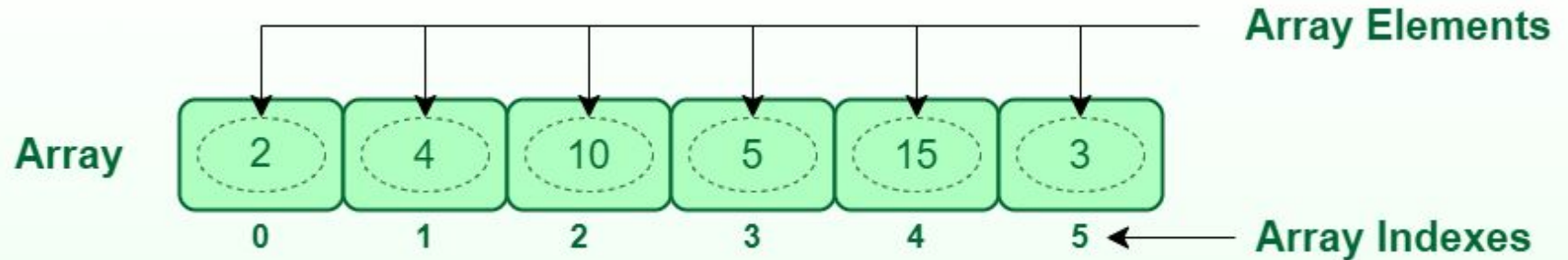
Arrays

جداول



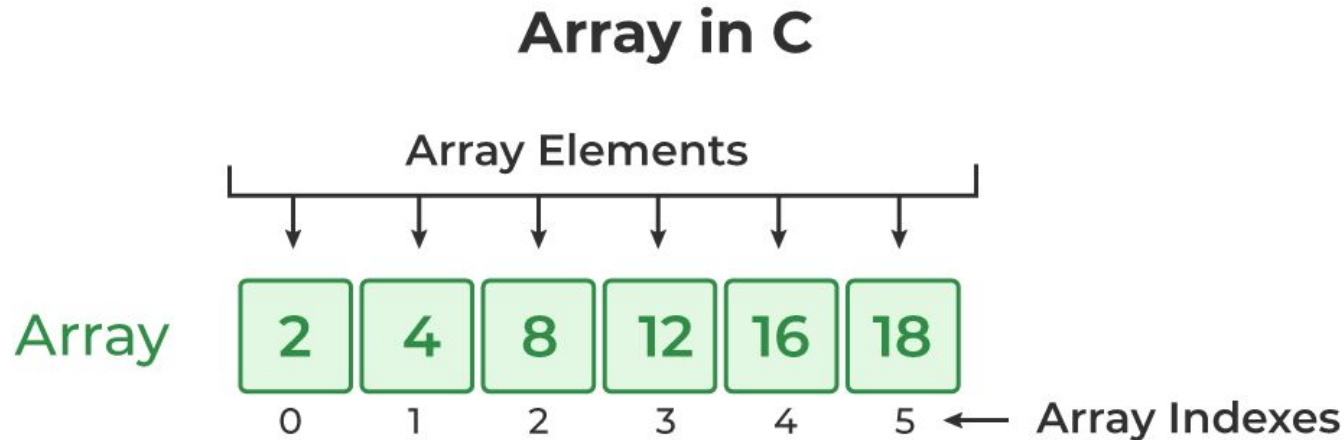
Arrays

Arrays are data structures in programming used to store a collection of elements of the same type



Arrays 1-D

One-dimensional arrays, also known as **simple arrays** or **vectors**, are arrays that store elements in a **single row** or a **single column**



Arrays 1-D

Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin var arr : array[1..5] of integer arr[1] := 10 End</pre>	<pre>#include <stdio.h> int main() { int arr[5]; arr[0] = 10; return 0; }</pre>

Arrays 1-D

```
#include <stdio.h>
```

```
int main() {  
    int myNumbers[] = {25, 50, 75, 100};  
    int i;  
  
    for (i = 0; i < 4; i++) {  
        printf("%d\n", myNumbers[i]);  
    }  
  
    return 0;  
}
```



25
50
75
100

Arrays 1-D

```
#include<stdio.h>
void main()
{
    int arr[5]={7,23,34,12,9}, i, sum = 0;

    //add all elements to the variable sum.
    for(i = 0; i < 5; i++)
        sum = sum + arr[i];

    //print the result
    printf("Sum of the array = %d\n",sum);
}
```



Sum of the array = 85

Arrays 1-D

```
#include<stdio.h>
void main()
{
    int arr[5]={12,24,31,18,19};
    int odd_count = 0, even_count = 0, i;

    for(i = 0; i < 5; i++)
    {
        if(arr[i] % 2 == 1)
            odd_count++;
        else
            even_count++;
    }

    //print the result
    printf("Odd number count = %d\n",odd_count);
    printf("Even number count = %d\n",even_count);
}
```



Odd number count = 2
Even number count = 3

Arrays 2-D

- Two-dimensional arrays, also known as matrices or grids
- Are arrays that store elements in a tabular form with rows and columns with the same type (int, float, char,)

	Column 0	Column 1	Column 2
Row 0	x[0][0]	x[0][1]	x[0][2]
Row 1	x[1][0]	x[1][1]	x[1][2]
Row 2	x[2][0]	x[2][1]	x[2][2]

Arrays 2-D

Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin var arr : array[1..3, 1..3] of integer arr[1, 1] := 10 End</pre>	<pre>#include <stdio.h> int main() { int arr[3][3]; arr[0][0] = 10; return 0; }</pre>

Arrays 2-D

```
#include <stdio.h>

void main() {

    // declare a 2*3 matrix (2-D array) and initialize it
    int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };

    printf("%d", matrix[0][0]); // output 1

    // edit
    matrix[0][0] = 9;
    printf("%d", matrix[0][0]); // Now outputs 9 instead of 1

}
```

Loop through an Array 2-D

```
#include <stdio.h>

int main() {
    int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };

    int i, j;
    for (i = 0; i < 2; i++) {
        for (j = 0; j < 3; j++) {
            printf("%d\n", matrix[i][j]);
        }
    }

    return 0;
}
```



1
4
2
3
6
8



Thank
You!

